



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

CHARCOAL ADSORBERS (D-2204A/B/C)

Docu. No. : 14780-8120-PS-D2204
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
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TITLE :

CHARCOAL ADSORBERS

(D-2204A/B/C)

DISCIPLINE : **PROCESS**
DOCUMENT NO. : **14780-8120-PS-D2204**
PROJECT NAME : **PTT Phenol Train II Project**
PROJECT NO. : **120117**
OWNER : **PTT Phenol Company Limited (PPCL)**

Z1	MAY 10, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	MAY 22, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A2	DEC 11, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
A1	OCT 2, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
O1	JUN 27, 2013	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 10, 2016	MAY 10, 2016	-	MAY 10, 2016	

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VESSELS

PROJECT SPECIFICATION

963765 — 301

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	27APR12	TS	HDT				

General Notes

- A. Vessels shall be constructed in accordance with the following unless otherwise specified on the individual vessel specification sheet.

Standards:

UOP Standard Specification 3-11

- B. Although the resulting inside diameter may be slightly different than specified, vessels without trays or packing, may be fabricated from pipe.
- C. The dimensioned location of some of the vessel level nozzles may be noted as min (minimum) or max (maximum). It is the responsibility of the contractor to adjust these dimensions to suit the specific level requirements specified on the UOP Piping and Instrument Diagram(s) and on UOP Standard Drawing 8-121. When additional nozzles are required for level instrumentation, the contractor shall specify them consistent with the size, class and facing requirements specified in the UOP Project Specification 301-Vessels and on the UOP Piping and Instrument Diagram(s).
- D. When system of measurement is metric, all dimensions are in millimeters unless noted otherwise.
- E. Contractor shall not locate TI thermowells in a downcomer. Contractor shall also ensure that TI thermowells are located to avoid interference with downcomers, weirs, distributors, collectors, baffles, chimneys, traps and all other vessel internals.
- F. Suppliers (or equal):

Spray Nozzle:

BETE Fog Nozzle Inc.
www.bete.com

Profile Wire Screens:

Johnson Screens
www.johnsonscreens.com

Nagaoka International Corporation
www.nagaokajapan.co.jp

Other specifications
referenced in this
specification:

UOP Standard Specification: 3-11, 3-15, 3-35, 9-11

UOP Standard Drawing: 3-115, 3-116, 3-122, 3-123, 3-144, 3-183, 3-190, 3-264, 3-266, 3-313, 3-322, 8-121

UOP Project Specification: 963765-106, 963765-307

R
E
V

VESSELS

SERVICE	CHARCOAL ADSORBERS (THREE REQUIRED)	ITEM NO	D-2204A/B/C
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Design	INT	3.5 kg/cm2(g)	@ 195° C
Conditions	EXT	FULL VACUUM	@ 250° C
at Top		Metal Temperature(Min)	15° C
Operating	NORMAL	0.05 kg/cm2(g)	@ 16° C
Conditions	REGENERATION	0.2 kg/cm2(g)	@ 119° C

Radiograph	MR
Postweld Heat Treat	MR

Material	Corrosion Allowance mm (Min)
SA 240 TYPE 304L (NOTE 1)	1.5

This Vessel is in the Following Service(s):

Hydrogen	NO
Hydrofluoric (HF) Acid	NO
Hydrofluoric (HF) Acid (with Resid. Elem. Control)	NO
Wet Hydrogen Sulfide (H2S)	NO
Severe Cyclic (NOTE 9)	YES
Caustic	NO
Carbonate	NO
Amine	NO
Note	

Vessel Heads	
Vessel	2:1 ELLIPTICAL

Accessories Applied by Fabricator	
Ladder & Platform Clips	AS REQUIRED
Insulation Clips & Rings	H (80) YES
Vessel Support	YES

Nozzles and Manways			
Mark	No	Size Inches	Service
AA	1	30 ID	MANWAY
AI	1	30	VAPOR INLET (NOTE 3)
AM	1	30 ID	MANWAY
AN	1	30 ID	MANWAY
AO	1	30 ID	MANWAY
AS	1	18	STEAM INLET
AX	3	2	TI (2)
AY	1	2	TI (2)
AZ	1	2	TI (2)
BA	1	6	RELIEF
BB	1	30 ID	MANWAY
BC	1	24 ID	MANWAY
BG	1	3	QUENCH (NNF)
BH	1	4	CONDENSATE OUTLET
BN	1	30	OUTLET (NOTES 3 & 7)
BO	1	30 ID	MANWAY
BP	2	2	PDI (2)
BV	1	2	VENT
BX	2	2	TI (2)
BY	1	2	TI (2)
BZ	1	2	TI (2)

Class	- ASME CL150
	(2) ASME CL300
Facing	- RAISED FACE

Normal Liquid Level	= N/A
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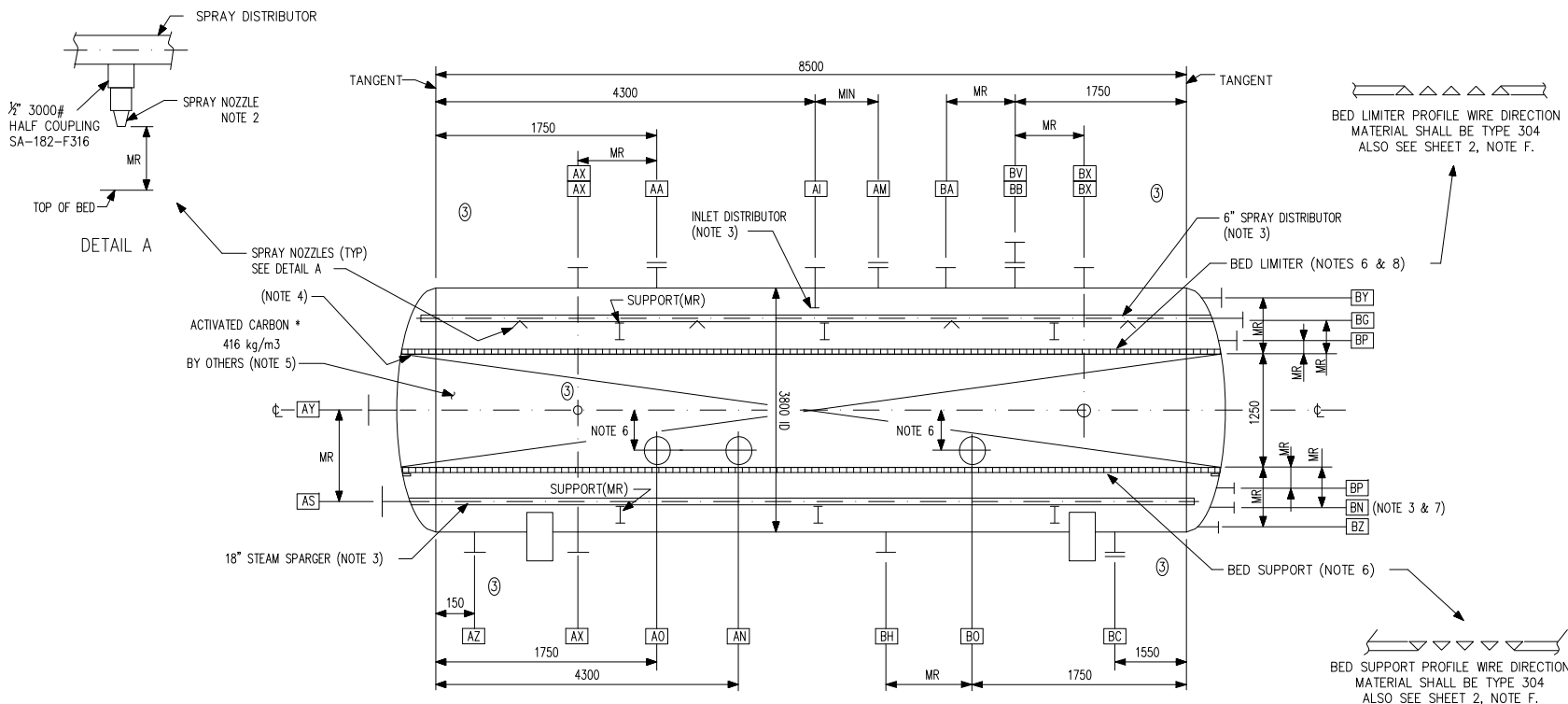
Specific Gravity	= 1.0 (DESIGN LIQUID FULL)
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- NOTES: 1. VESSEL TO BE DESIGNED AND FABRICATED IN ACCORDANCE WITH UOP STANDARD SPEC 3-15.
2. USE 150 FULL CONE SPRAY NOZZLE, BETE FOG TYPE "F" TF 28-150. FOR SUGGESTED SUPPLIER SEE NOTE "F" ON SHEET 2.
3. INLET DISTRIBUTOR, SPRAY DISTRIBUTOR, STEAM SPARGER AND PROCESS OUTLET COLLECTOR PER SYSTEM DESIGNER RECOMMENDATION. TOTAL PRESSURE DROP FROM INLET NOZZLE "AI" TO OUTLET NOZZLE "BN", INCLUDING BED, SHALL NOT EXCEED 0.02 kg/cm2.
4. PROVIDE 150 MM CLEARANCE BETWEEN EXPECTED TOP OF THE CHARCOAL BED AND BED LIMITER.
5. "F" EXTRUDATE PER CHARCOAL SUPPLIER RECOMMENDATION.
6. FOR BED SUPPORT, BED LIMITER AND MANWAY LOCATION SEE UOP STANDARD DRAWING 3-116, EXCEPT BAFFLE OMITTED.
7. NOZZLE "BN" SHALL NOT BE SUBMERGED IN LIQUID.
8. PROVIDE BOLTED ACCESS SECTION (MANWAY) IN BED LIMITER. ALIGNED WITH MANWAYS "AA", "AM" AND "BB".
9. VESSEL CYCLES BETWEEN 16°C, 0.02 kg/cm2(g) AND 120°C, 0.16 kg/cm2(g) EVERY 6 HOURS.

PROJECT SPECIFICATION

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REV	DATE	BY	APP'D	REV	DATE	BY	APP'D
0	27APR12	TS	HDT				
1	03JUL12	JG	HDT				
2	07SEP12	TS	HDT				
3	03SEP13	TS	HDT				



Drawings Referred to in this Specification

3-116						
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